

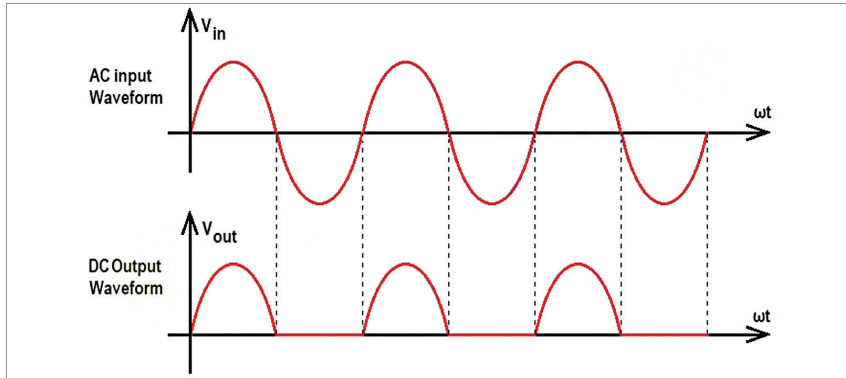


Fig. 3: Input and output waveform of half-wave rectifier without filter

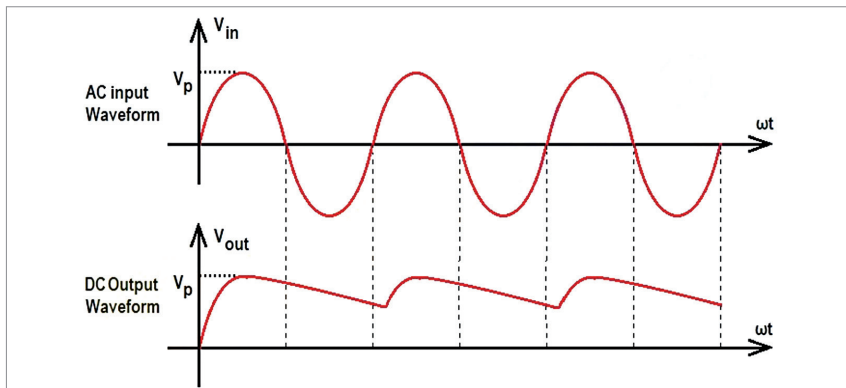


Fig. 4: Input and output waveforms of half-wave rectifier with filter

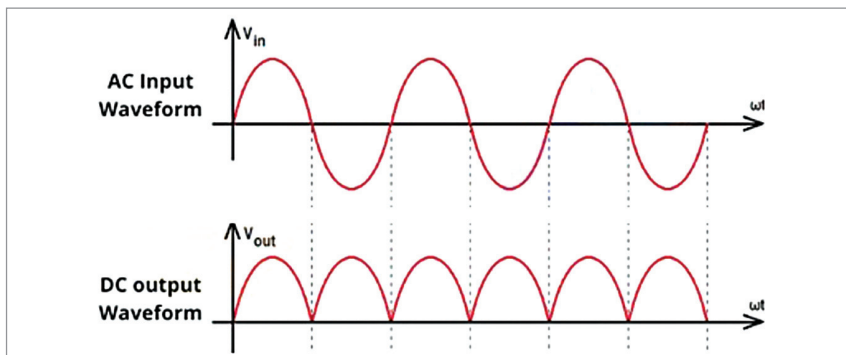


Fig. 5: Input and output waveforms of full-wave rectifier without filter

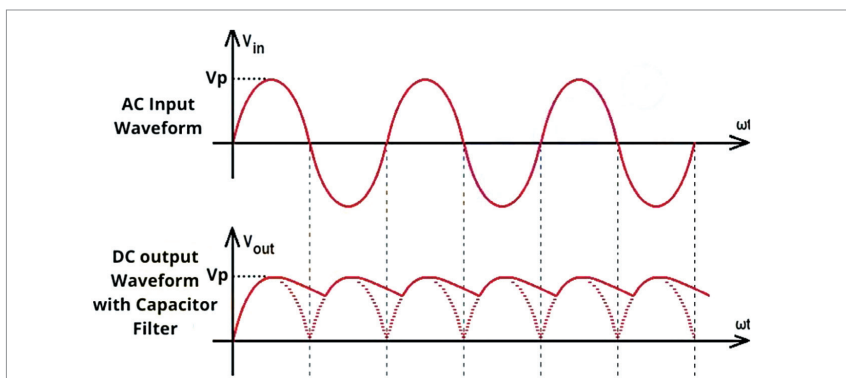


Fig. 6: Input and output waveforms of full-wave rectifier with filter

terminal of capacitor C1 is connected to centre-tap on secondary side of the transformer. The output waveform can be observed on an oscilloscope connected across resistor R1.

The centre-tap transformer is used to step down the supply voltage from 230V to 9V-0-9V.

The two 1N4007 diodes (D1 and D2) and 100μF, 25V electrolytic capacitor (C1) are used to get different output waveforms.

Construction and testing

Solder all the components on a Veroboard and upload the source code (Study.ino) on Arduino Uno board using Arduino IDE. Place the CRO probes across resistor R1.

When the circuit is switched on, you get half-wave output without filter capacitor C1, as shown in Fig. 3. Next, the program automatically outputs half-wave output, as shown in Fig. 4. This is followed by full-wave output without filter, as shown in Fig. 5, and full-wave output with filter, as shown in Fig. 6. The program keeps repeating in this sequence to output different waveforms.

As can be seen from Fig. 3 through Fig. 6, the waveforms available include the input and output voltages for comparison. The DC output obtained from pulsating DC input can be smoothed by increasing the value of filter capacitor C1. You can also connect a voltmeter across the diodes and centre-tap of transformer to measure the output voltage. **EFY**



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